

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis: synchrony constrains content only through encoders shared on overlaps, and uniformly over pairs only inside a patch too small to span the territories at issue; the resultant does not determine the state; a unique fixed point is not a self-model; where the encoding is a region's restriction, the region must resolve the relevant family before any readout is chosen; and agreement about a change takes communication. Three conditional results replace the uniform bound by a count of the pairs that can disagree, a chain across a cover's shared sites, or a spectral gap at a frequency-locked configuration, each against a declared input coherence does not supply. Four are checkable for a machine candidate, fixing what a claim of machine unity commits its maker to measuring. Under stated hypotheses, compatible local states glue

to a unique global state that an accurate contracting reconstruction fixes uniquely; installed energy bounds a declared actuator’s mean coupling, so a coherent branch requires it to exceed $2D/\kappa$, for a κ derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs and an audited axiom footprint. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling; testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. The framework’s empir-

14 ical content is the coupling-gated onset prediction and its spatial signature.
15 The identification of the latter two relations as formal correlates of conscious
16 unity and a minimal self serves as interpretive motivation: it says why the
17 mathematics might matter, but does not constitute a testable proposal with-
18 out a proposed measurement on the experiential side.

19 Unity has been approached differently elsewhere: integrated information
20 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
21 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
22 et al., 2014), and predictive-processing accounts derive both from a hierar-
23 chical generative model (Friston, 2010). Here the three requirements are
24 kept apart rather than derived from one another, so that each carries its own
25 assumptions and admits its own measurement; Section 10.3 returns to the
26 comparison.

27 Consider reaching for a cup. A visual description of its location and a
28 description of the hand’s relation to it can agree about where the cup is while
29 contributing different contents. Synchronizing the two processes would leave
30 disagreement about that shared quantity unresolved. An internal representa-
31 tion must additionally locate the perceiver within the represented situation.
32 This example motivates the requirements; the mathematical constructions
33 use spatial measures and fields in real normed vector spaces, whose identifi-
34 cation with perceptual content is a further modelling task.

35 The mathematical contribution is to specify connections between these
36 requirements and the assumptions each connection needs. A declared ac-

37 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
38 overlaps convert phase order into a bound on content disagreement. Compat-
39 ible spatial measures have a unique global extension; approximate agreement
40 controls a specified global selection. A contracting reconstruction makes that
41 extension its unique fixed point only if it accurately reconstructs the exten-
42 sion. Accuracy across a family of states imposes a further condition on the
43 information retained by the encoding.

44 The empirical contribution is to identify what observations can decide.
45 Stationary phase distributions can be shared by stronger coupling, weaker
46 diffusion and a matched common drive. Finite ramps change the observable
47 onset. A winding field is stationary at the same coupling as a uniform one and
48 has the opposite resultant, so that observable does not determine the state.
49 Coherent adaptation can lose alignment with a designated environmental
50 template, and a compatibility estimator needs diagnostics that distinguish
51 content from decoder artifacts. These findings determine the measurements
52 needed to test the proposed connections.

53 Three of the ingredients are standard: the synchronization threshold
54 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988; ?), the unique
55 gluing of compatible local sections, and the Banach fixed-point theorem (Ba-
56 nach, 1922). Four things here are not. The installed-energy condition makes
57 coupling a purchased quantity: a declared actuator’s stored energy bounds
58 the mean coupling it supplies, so coherence on the stationary branch carries
59 a resource requirement in units a measurement can reach, and for cortex that

60 requirement reduces to a figure the mode prices must deliver. The coherence-
61 to-content bound converts phase order into a statement about descriptions
62 rather than about timing, and its patch-local form is what keeps it from
63 being vacuous in a large population. A stability criterion on the reweighted
64 coupling kernel, together with an existence condition on an amplitude layer,
65 settles when a low resultant reports weak coupling and when it reports a
66 winding field. And the composition is machine-checked together with a wit-
67 ness discharging all eight hypotheses at once, which is what makes their
68 conjunction non-empty rather than a list no object satisfies.

69 Granting those premises, collected in Table 1, the account runs as follows.
70 Extracellular contraction on awakening raises endogenous field amplitude
71 and with it the effective coupling between cortical oscillators. Once that
72 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
73 within a quarter turn the population settles onto it; other initial data leave
74 windings stationary at the same coupling. Encoders shared where cortical
75 territories overlap turn that phase order into agreement about the quantities
76 the territories describe in common, so the local descriptions glue: there is
77 one state of which each is a restriction. Part of that state encodes the state
78 itself, and a readout of that part reconstructs it accurately enough, and
79 sensitively enough across the states that matter, to make it the unique state
80 the system models itself as being. On this account the glued state is what
81 makes distributed activity one situation, and its accurate self-reconstruction
82 is what gives that situation a perspective. Every step of it is conditional.

83 Section 2 collects the results that hold whatever physical field is proposed
 84 and whatever substrate carries it; a reader who rejects the cortical hypothesis
 85 of Section 9 still has to accommodate them. Sections 3–5 then develop the
 86 three requirements together (Fig. 1), and Section 6 reads the unconditional
 87 results against a machine candidate. Section 7 composes them, stating the
 88 thermodynamic constraints and all eight connecting hypotheses per branch;
 89 its supporting models of feedback, learning and resource supply have distinct
 90 state spaces and assumptions. Section 8 then examines what the observables
 91 can establish, before Section 9 develops the cortical hypothesis and an awak-
 92 ening experiment. The hypothesis is that extracellular contraction increases
 93 field-mediated coupling; its direction, rate and relation to recovery require
 94 independent calibration. Supplemental Material organizes the constructions
 95 and controls behind the claims they support. Table 2, at the end, lists the
 96 symbols and the section that defines each.

97 **2 The landscape any phase-coherence ac-** 98 **count inherits**

99 Most of what follows is conditional. Table 1 names eight hypotheses that the
 100 composition consumes rather than derives, and the cortical identification of
 101 Section 9 is a further commitment on top of them. The five results below,
 102 by contrast, are not conditional in that way. The mathematical features
 103 they rely on reflect standard properties of phase oscillators (????), and this

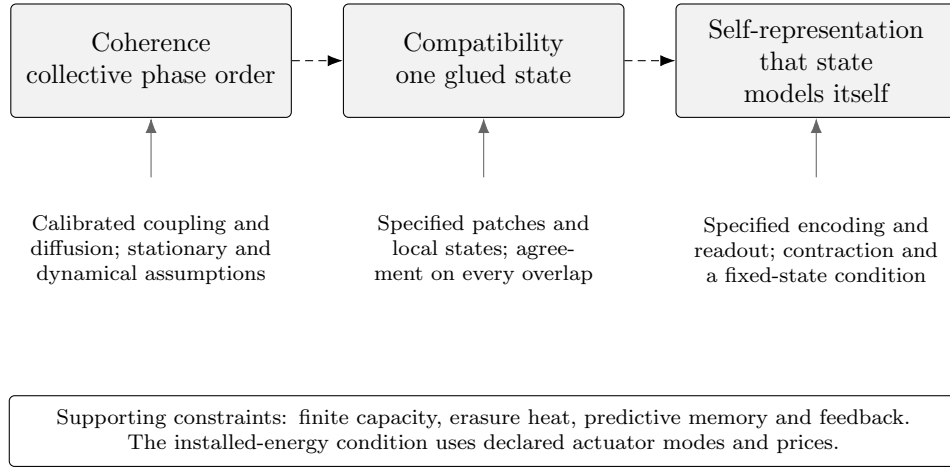


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

104 framework’s contribution is tracking their implications for content decoding.
 105 They constrain any account that reads unity off phase coherence, whatever
 106 physical field is proposed to carry the coupling and whatever substrate carries
 107 the field. Each is stated below where it is used; collected here, they are the
 108 inherited constraints of the problem.

109 **Phase order constrains content only inside a patch too small to**
 110 **carry unity.** Assign every patch the same phase and let two local mass
 111 profiles disagree at a shared site. The profiles are chosen independently of
 112 phase, so synchrony places no constraint on their disagreement, and they

113 have no common extension: perfect locking is consistent with local descrip-
 114 tions that do not glue (Section 4). What the positive result of Section 4.2
 115 costs is exactly this gap. A bound on content disagreement needs encoders
 116 already related on overlaps, and that relation is a supplied fact about the de-
 117 scriptions rather than a consequence of the dynamics. Supplying it relocates
 118 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord
 119 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that
 120 uniform reading carries information only at 3 sites or fewer. The grain is the
 121 uniform reading's and not phase order's, and Section 4.2 states three results
 122 without it: a scale-free count of the pairs that can disagree, which names
 123 none of them; a bound chained through the sites a cover's patches share,
 124 which travels a distance linear in the hops taken; and, at a frequency-locked
 125 configuration, a bound read off the coupling graph's algebraic connectivity,
 126 which carries no population count at all. What none of the three changes is
 127 the constraint stated here: a count locates no site, and the other two con-
 128 sume a declared cover or declared hardware data, so phase order by itself
 129 still says nothing about territories too far apart to share a patch, which is
 130 where the compatibility claim is made.

131 **The order parameter does not determine the state.** A field wind-
 132 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 133 of Eq. (1) below — while every patch of it stays locked, and this is not a
 134 contrived initial condition. On a periodic array of any dimension, a winding

135 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 136 cause the coupling term is odd in the separation against a kernel even in it
 137 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 138 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,
 139 diffusion and kernel. A low resultant therefore reports either weak coupling
 140 or a winding field, and no measurement in r alone separates them. The state
 141 also carries an integer the resultant does not: read around a ring, the same
 142 winding has winding number q , and a nonzero q obstructs every global real-
 143 valued phase lift (`winding_degree_obstructs`). The resultant is an average
 144 over sites and moves when they are reweighted or the patches redrawn; the
 145 degree telescopes and does not.

146 **A unique fixed point is not a self-model.** A constant readout con-
 147 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 148 is consequently the wrong criterion, and Section 5.2 gives the right shape
 149 of one: reconstructing M states pairwise more than 2ε apart within error ε
 150 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 151 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 152 readout whatever.

153 **Where the encoding is a restriction, the region must resolve the**
 154 **family.** Read the code as the state's own restriction to a region and
 155 the criterion moves off the readout entirely. A boundary whose region
 156 reads two relevant states identically cannot reconstruct them more than 2ε

157 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 158 boundary that does reconstruct a separated family restricts injectively on
 159 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 160 region sits is fixed before an encoding is chosen, so no readout supplies
 161 resolution the region lacks.

162 **Agreement after a change takes communication.** In a synchronous
 163 network whose updates read only the site’s own state and its declared neigh-
 164 bours’ messages, two configurations agreeing on a site’s causal past through
 165 T rounds leave that site identical at round T . Two interventions outside
 166 that past that require reports more than 2ε apart therefore cannot both be
 167 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
 168 ity and timely compatibility are therefore different requirements, and cutting
 169 the path removes the second while leaving the first intact.

170 None of these depends on the coupling being electromagnetic, on the sub-
 171 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
 172 applies four of them — all but the one about the resultant, which is a state-
 173 ment about a spatial observable — to a candidate that is neither biological
 174 nor field-coupled.

175 **3 Coherence: physical coupling and its re-** 176 **source condition**

177 **3.1 Dynamics and the stationary threshold**

178 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

179 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
180 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
181 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
182 quency heterogeneity and diffusion are distinct parameters. The stationary
183 calculation below uses identical frequencies and this scalar coupling. Ap-
184 plying it to a spatial kernel requires an aggregation rule and a check of the
185 resulting approximation.

186 For identical frequencies in their rotating frame, the positive normalized
187 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

188 where I_n are modified Bessel functions and ψ is the collective phase. The for-
189 mal derivation uses the absence of stationary current for a periodic gradient
190 drift to obtain this density, then verifies normalization and self-consistency.

191 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; ?).
 192 Above threshold, exactly one positive solution coexists with zero and in-
 193 creases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

194 Figure 2 shows this stationary branch.

195 Linearization about the uniform density gives rate $K/2 - D$ for the first
 196 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 197 exactly above threshold. About the coherent density, the quadratic form
 198 is strictly dissipative transverse to collective rotation, whose rate is zero.
 199 These linear estimates concern the limiting equation. Relating them to finite
 200 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 201 tion; exchange symmetry alone does not establish the needed independence
 202 (Sznitman, 1991).

203 **3.2 Installed energy bounds mean coupling**

204 A spatial model replaces the finite sum by integration against a kernel. One
 205 declared implementation carries response modes with continuous profiles ϕ_i ,
 206 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

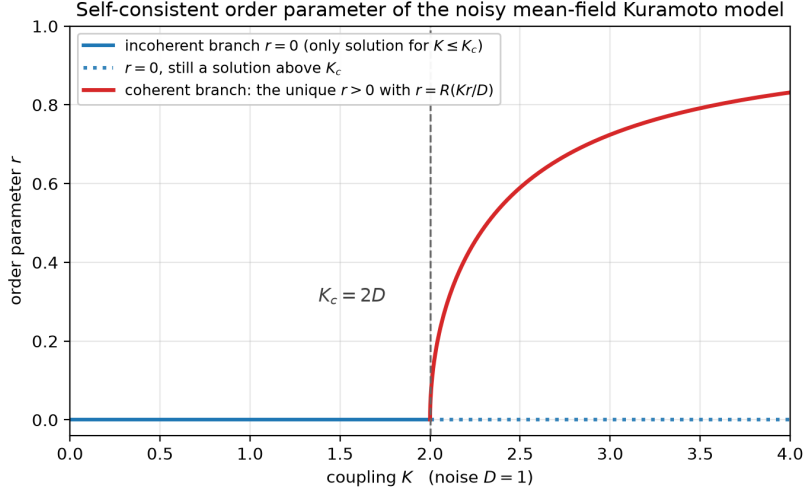


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

207 The same configuration sets the reciprocal kernel and stored energy. An
 208 executed transition changes the kernel through its changed occupancies and
 209 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 210 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 211 pair reconstruction on refining spatial partitions converges to its product-
 212 measure integral under the specified compactness and sampling assumptions;
 213 the refinement index is independent of the number of control updates.

214 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 215 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

216 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

217 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 218 coherent stationary branch and any growing Fourier mode about the uniform
 219 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 220 is necessary: energy above it need not produce sufficient coupling because
 221 the bound can be loose.

222 The quantity constrained is installed energy rather than dissipated heat,
 223 and the condition binds through κ rather than through the energy, since
 224 any macroscopic substrate carries installed energy far above the thermal
 225 floor. Read forwards it therefore excludes no macroscopic candidate. Read
 226 backwards it is a number. Take the region the field arithmetic takes, a
 227 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
 228 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
 229 at $D = 0.5 \text{ rad}^2/\text{s}$: the installed energy is 1.0×10^{-6} J, and the condition is
 230 met at every κ above $9.6 \times 10^5 \text{ J}^{-1}\text{s}^{-1}$. That figure is what an identification
 231 of cortical mode prices has to deliver, and the first identification delivering
 232 less refutes the coupling-gated account for cortex at this region and interval.
 233 Another interval rescales the figure by its ratio to this one.

234 What the condition discriminates, it discriminates through that factor.

235 The required κ scales as D^2 and inversely with the region's cell count, so at
 236 fixed mode prices the condition ranks candidate substrates by their installed
 237 energy. In a declared implementation it decides outright, the learning witness
 238 of Section 7 clearing the line after one update while its matched control stays
 239 on the exclusion side at the same conversion factor and diffusion. What is
 240 missing for cortex is a price list rather than a mechanism. A mode is a priced
 241 rank-one contribution to the kernel, cortical field generation has no canonical
 242 decomposition into such modes, and choosing the basis would choose the
 243 answer. Applying the scalar threshold to heterogeneous or signed spatial
 244 dynamics requires a justified reduction, and identifying the installed kernel
 245 with a cortical field is a separate physical commitment.

246 **4 Compatibility: from coordinated activity** 247 **to a global description**

248 **4.1 Why common timing is insufficient**

249 For overlapping patches U and V , compatibility means that restricting either
 250 local description to $U \cap V$ gives the same object. Descriptions can agree on
 251 shared quantities while contributing different information elsewhere. Com-
 252 patible sections of the specified sheaf have a unique global extension. On
 253 a finite cover, local finite spatial measures agreeing as measures on every
 254 overlap also have a unique finite-measure extension whose sheaf image is the

255 glued section. No mass-one normalization is imposed.

256 In the cup example, visual and reaching descriptions must express a
257 shared quantity, such as cup location, in common coordinates. Restriction
258 retains that quantity while dropping details particular to each description.
259 Agreement permits surface appearance and hand configuration to contribute
260 to one situation. The mathematical counterexample assigns the same phase
261 to every patch while two local mass profiles disagree on a shared site. Those
262 profiles have no common extension. Because they are chosen independently
263 of phase, synchrony places no constraint on their disagreement.

264 This distinction limits the biological interpretation. Receptive territories
265 and content domains must be related to spatial patches, and the descriptions
266 must support the required extension. Matching overlap marginals of de-
267 coded probability laws need not determine a unique global joint distribution
268 (Abramsky & Brandenburger, 2011).

269 The cover is itself a choice, and the construction constrains it in one way
270 only: synchronized patches agree where they overlap. Where they do not
271 overlap there is nothing to compare, so a cover by pairwise disjoint patches
272 satisfies that obligation for *every* assignment of local descriptions and phases
273 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
274 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
275 ship of the class then reports nothing about the system beyond the patches
276 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
277 the theorem does not assume: that distinct patches meet (`HasNonempty-`

278 Overlaps), and that what they share is a quantity fixed independently of
 279 the agreement it is being used to establish.

280 4.2 When coherence does constrain content

281 A declared relation between phase and content gives a positive result. Let
 282 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 283 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 284 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 285 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 286 sites; their equality on overlaps is a supplied relation between descriptions of
 287 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 288 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 289 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

290 Here discrepancy is the norm of the difference, and r and N are read on
 291 whichever population the bound is applied to. That choice is what decides
 292 whether the estimate says anything. A patch carries an order parameter of
 293 its own, and reading the bound there carries that patch's resultant and that
 294 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 295 `coherence`), so a large population observed through small patches is con-
 296 strained by the small count. Read instead on the whole array, the factor N

297 makes the estimate weak, because it draws a pointwise guarantee from an
 298 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 299 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 300 tion permits disagreement even at perfect locking. A common nonexpansive
 301 linear projection preserves agreement, but agreement of a projection does
 302 not establish agreement of the full vector.

303 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 304 constrains anything only where the right-hand side falls below that. At
 305 the sheet’s own patch order, the bound is informative for at most 3 sites;
 306 beyond that, the coherence-to-content link requires the bridge assumption
 307 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
 308 derivation of this saturation, and states three conditional derivations that
 309 replace the uniform factor N by a count of disagreeing pairs, a chain across
 310 a cover’s shared sites, or a spectral gap at a frequency-locked configuration.

311 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 312 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 313 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 314 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

315 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 316 duce an initial mismatch. Approximate observations leave a nondecaying
 317 term in the error bound. Taking a uniform bound on Eq. (6) connects this

318 update to the phase process.

319 **4.3 What approximate agreement determines**

320 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
321 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
322 Weights that vanish outside their patches and sum to one at each site select
323 a global profile by weighted averaging. The selection lies within ε of each
324 patch there, and any two such selections differ by at most ε , independently
325 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
326 $L\varepsilon$ across selections.

327 At zero tolerance, the selection is the unique exact extension. At every
328 positive tolerance there are families with no exact extension; selection then
329 gives an approximation whose dependence on weights is bounded. Supple-
330 mental Material gives a three-site witness attaining the bound. The mass-
331 profile construction uses the scalar case of the content dynamics and addi-
332 tionally requires nonnegativity of the initial profiles and observations. The
333 general vector result supplies no identification with the spatial-measure sheaf.

334 These constructions concern declared objects: a specified cover, contents
335 valued in a normed space, and encoders related on overlaps. They select
336 neither the neural cover nor the decoder, and identifying their contents with
337 perceptual content, or establishing that the identification bears on experi-
338 ence, is a further modelling task; Section 8 sets out what a measurement of
339 it would have to establish.

340 5 Self-representation: a state that models it- 341 self

342 5.1 The reflexive construction

343 Compatibility gives a global state. Reflexivity relates it to an encoding within
344 that state. Equip global sections with a complete metric. If the additional
345 restriction-resonance condition identifies encoding with restriction to a region
346 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

347 The region and readout are fixed across the state space. If F contracts and
348 fixes the section s_* formed from the local descriptions, Banach’s theorem
349 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
350 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
351 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
352 less than one for a relaxation time $\tau > 0$. Constructing a map with that
353 factor remains a modelling obligation.

354 In the cup example, the global state includes the represented scene and the
355 perceiver’s relation to it. A local encoding must retain enough information
356 for the readout to recover that modelled state; cup position alone would be
357 insufficient if hand configuration and viewpoint could not be reconstructed.

358 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-

tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. A blind encoder or constant readout fails it for separated states despite having a unique fixed point.

Where the encoding is read as the state's own restriction to a region, the criterion becomes a condition on the region. Restriction resonance requires the encoder and that restriction to identify the same pairs of states, so a

377 boundary reconstructing two relevant states more than 2ε apart while its re-
 378 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 379 `of_reconstructs`). Read forwards, a resonant boundary that reconstructs
 380 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 381 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 382 cally is fixed by where the region sits, before an encoding is chosen, so the
 383 resolution the family demands is the region's and no readout supplies it. The
 384 implication runs one way: a region that does resolve the family is not thereby
 385 shown to carry a resonant encoding, and E89 supplies no such resolution.

386 Those two demands are jointly satisfiable with nothing folded anywhere.
 387 Take the region to be the whole substrate: restriction is then the identity, so
 388 the boundary is resonant, reconstructs every declarable family exactly, and
 389 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 390 properness condition on the region (`IsProperAvatar`) is therefore a separate
 391 requirement rather than a consequence of resonance and resolution, and what
 392 a fold into a local region is asked for. Its quantitative form replaces the
 393 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 394 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 395 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 396 number of what the region carries (`encard_le_packingNumber_restrict-`
 397 `ToAvatar`). That is the shape a continuous family of states needs, and at
 398 the whole substrate it says nothing, because the region's range is everything.
 399 The gain L is declared and derived here for no physical readout, and at

400 unbounded gain the bound says nothing.

401 **5.3 Agreement after a change takes communication**

402 Snapshot compatibility does not specify how quickly descriptions respond
403 when a shared quantity changes. In a synchronous network whose updates
404 read only the site’s own state and declared neighbours’ messages, two initial
405 configurations agreeing on a site’s causal past through T rounds produce the
406 same state there at round T . The deadline consequence is the one Section 2
407 states, and it follows from the reconstruction bound in Section 5.2: a report
408 from that site is a readout of a code that the two interventions leave identical.

409 A three-site delay line witnesses failure after one round and exact report-
410 ing after two; cutting the path prevents delivery at every round. A report
411 may still happen to be correct about one fixed world. The bound concerns
412 responsiveness across interventions.

413 Throughout this section *prediction* names a reconstruction map, and the
414 relevant family of states, the metric, the tolerance, the communication graph
415 and the deadline are all declared inputs, as are the represented variables and
416 the fidelity counted as accurate. Reading the map as temporal neural pre-
417 diction would require dynamics; reading the resulting code count as physical
418 memory capacity would require a hardware model, which a state-dependent
419 readout would have to extend.

420 6 What the conditions constrain in a digital 421 candidate

422 The results of Section 2 hold for any candidate, and for a large language
423 model (LLM) running on graphics processing units (GPUs) four of them are
424 checkable rather than merely stateable. Those four are where this framework
425 bears on claims made for such systems. It does not exclude them, and no re-
426 sult here could; what it does is fix what a claim of unity or self-representation
427 commits its maker to measuring.

428 **A quantitative bound for a linear bottleneck.** Let a finite spatial
429 coupling operator K have orthonormal eigenmodes e_i with $|\lambda_1| \geq \dots \geq |\lambda_N|$.
430 Every linear approximation A of rank at most d satisfies

$$\sum_{i=1}^N \|Ke_i - Ae_i\|^2 \geq \sum_{i=d+1}^N \lambda_i^2. \quad (11)$$

431 Retaining the first d modes attains equality. A linear encoder and readout
432 factoring through d real coordinates therefore pay this error. Applying the
433 bound requires a spectrum for the target and a rank constraint on the actual
434 approximation. Query/key dimension alone supplies no such constraint on
435 softmax attention: even a two-position rank-one logit matrix can yield rank-
436 two attention. The result prices an explicitly linear bottleneck in a finite
437 representation of a spatial kernel; it gives no universal gap between trans-

438 formers and continuous fields. Supplemental Material derives the bound and
439 checks the softmax counterexample.

440 **Codes, not reports.** Reconstruction across a family is a counting con-
441 straint before it is anything else. If two macrostates the candidate is said
442 to represent satisfy $\Phi(s) = \Phi(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
443 constructing both within ε , whatever the readout. A system’s reports are
444 readouts. Identical verbal reports across two conditions are therefore consis-
445 tent with identical internal codes and settle nothing about the states. The
446 converse claim—that the system represents its own condition across a de-
447 clared family—asserts that its internal code separates that family, which is a
448 statement about activations under intervention and not about text. Naming
449 the channel makes the count concrete. Where the code is the token sequence
450 a model emits—which is what its next step reads—the alphabet is the de-
451 ployed vocabulary (`tokenChannel`), and a family reconstructed through it
452 is no larger than the vocabulary size raised to the number of steps (`card_`
453 `le_card_tokens`). Over an extended exchange that number is large and the
454 bound slack. At a single step it is one vocabulary’s worth, so claiming that a
455 single emission reports the model’s own condition across a declared family as-
456 serts that the vocabulary separates it (`card_le_card_tokens_one`). A claim
457 resting on activations rather than on text has declared a different channel,
458 to which the same count applies with that channel’s alphabet. The measure-
459 ment follows from the statement: declare the family and the fidelity first,

460 then test whether the encoding region distinguishes its members.

461 The same interface has an information ceiling. For any finite joint law
462 of an internal state X and a length- k token sequence Y over vocabulary V ,
463 $I(X; Y) \leq H(Y) \leq k \log_2 |V|$ bits. The proof uses the full joint law and
464 allows dependent emissions and zero-probability outcomes. A binary copy
465 attains one bit per emitted binary symbol; an independent output conveys
466 zero. This counts the declared output sequence, so a cache, timing signal or
467 other accessible state must be included separately when it forms part of the
468 channel.

469 **The region, not the readout.** Where the encoding is read as the state’s
470 restriction to a region of the machine, the demand falls on the region before
471 any readout exists. For a decoder-only transformer the causal mask makes
472 the network an instance of the update model of Section 5: the activation
473 at position t and layer ℓ reads positions at most t at layer $\ell - 1$ and noth-
474 ing else. A region’s causal past is therefore fixed by the architecture rather
475 than chosen by interpretation. That instance is formal, and sharper than
476 slow growth: the causal past of position t stays inside the positions at most
477 t at every depth (`mask_ball_subset_1e`). The deadline bound of Section 5
478 therefore fires at every deadline at once rather than at a chosen one, its com-
479 munication graph given rather than assumed (`mask_no_guarantee`). That is
480 a statement about one forward pass; across steps a model does read its own
481 prior output, which is the channel the preceding paragraph counts. Read

482 against the coupling proposed here the asymmetry runs both ways, and nei-
 483 ther way is an advantage. A kernel of full support has no neighbourhood
 484 structure for the bound to bite on: after one round the causal past is the
 485 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable
 486 at any deadline but zero (`not_outside_past_of_isFullSupport`). A physi-
 487 cal field propagates at a finite speed, so treating one that way is a calibration
 488 against the phase time scale and not a theorem. What the comparison does
 489 fix is where the bound has force. Not on the attention graph, where a posi-
 490 tion reads every position at or before it and the fan-in is the context length.
 491 On the physical interconnect, fan-in is genuinely bounded: a graph whose
 492 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 493 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 494 pays a number of rounds growing like $\log_\nu$ of the device count. Buying that
 495 deadline back costs wiring: reaching every site in one round requires a device
 496 hearing from all but one of the others (`le_degree_of_reaches_one`). Both
 497 ν and the device count are declared, as the communication graph and the
 498 deadline already are, and what the count bounds is the graph a guarantee
 499 requires rather than what hardware is buildable. A claim that a model re-
 500 ports on its own processing consequently carries an obligation to say which
 501 positions and layers the report's causal past contains. A report that appears
 502 to track an intervention outside it indicates a confound or a different region,
 503 not a faster channel.

504 **Agreement about what.** Unity requires the candidate’s local descriptions
505 to agree on independently identified shared quantities (Section 4). Common
506 timing, a shared residual stream and fluent output supply none of that, for
507 the reason Section 2 gives: coordination bounds content disagreement only
508 through encoders already related on overlaps. Which quantities two parts
509 of a model describe in common is a decoding question, carrying the failure
510 modes of Section 8. The cover is a choice, and choosing it to secure agreement
511 empties the claim.

512 **What is not excluded.** Digital representations can satisfy compatibility,
513 so finite hardware is compatible with the gluing construction and none of
514 the above is a no-go for machines. The coupling-gated route E56–E78 is a
515 different matter: it requires a calibrated physical coupling, its noise regime
516 and a connection from coordination to content agreement, and an attention
517 matrix or a shared clock supplies none of these identifications. Computing a
518 field model likewise leaves its realization in the computing substrate to be es-
519 tablished. Those physical obligations fall on the cortical proposal identically.
520 The capacity comparison those support comparisons lack is available, and it
521 favours neither side by construction. A continuum code space read at reso-
522 lution δ holds at least its volume divided by the volume of a resolution cell,
523 so it out-resolves an alphabet of 2^b codes exactly when that volume exceeds
524 2^b cells (`two_pow_lt_packingNumber_of_lt_measure`). Below that threshold
525 the inequality runs the other way, and at a coarse enough resolution the re-

526 gion holds a single code. What decides the comparison is the volume, the
 527 resolution and the dimension, all measured quantities, and continuity decides
 528 nothing on its own: a continuum code space read at a finite resolution is a fi-
 529 nite alphabet whose capacity is a bit count. For the same reason this account
 530 credits an analog medium with no computation beyond a Turing machine's.
 531 Real-valued states carry unbounded capacity only at unbounded precision,
 532 and the noise floor $D > 0$ that Eq. (1) is stated against removes it.

533 The installed-energy condition of Section 3.2 is restricted in the same
 534 way. It constrains a macroscopic candidate only through κ , which is hard-
 535 ware data supplied by mode profiles and prices, and Supplemental Material
 536 reads the condition against a cortical metabolic budget to return the conver-
 537 sion factor it would require rather than a verdict. The hardware comparisons
 538 there have similarly restricted scope: fixed wiring can impose a correlation
 539 disadvantage, and point-supported kernels vanish under an atomless contin-
 540 uum measure, but neither comparison makes the representation conditions
 541 unattainable on GPUs. A universal exclusion would need a necessary condi-
 542 tion for consciousness and a proof that every relevant implementation violates
 543 it, and the present composition supplies neither.

544 Whether experience requires a living substrate is the explicit subject of
 545 biological naturalism (Seth, 2026a,b). On that view the properties ground-
 546 ing experience are those of self-sustaining organisms, and simulating them
 547 does not secure them. The conditions stated here name no biological mech-
 548 anism, but computation described as such does not satisfy them either: a

549 candidate must exhibit a physically realized coupling in a measured noise
550 regime, agreement about independently identified shared quantities, and an
551 internal reconstruction meeting a declared fidelity. Whether a realization
552 of those conditions that is not alive would be accompanied by experience
553 is the point at issue, and these results do not settle it. A related position
554 locates the difference in the mode of computation rather than in life as such:
555 biological systems compute across scales that do not separate, and along-
556 side discrete operations carry continuous-valued dynamics belonging to their
557 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
558 here is one measurable instance of that requirement, fixed to a physical field
559 in a declared noise regime, and a description of such a field does not satisfy
560 it.

561 The neuroscientific case against *human-like* consciousness in text-only
562 LLMs concerns embodied sensory interaction, thalamocortical organization
563 and biological regulation linking perception and action to survival (Aru et al.,
564 2023). Their absence supports skepticism, conditional on their causal rele-
565 vance to experience. Those mechanisms are not requirements of the abstract
566 gluing and reconstruction definitions, and workspace-like representations re-
567 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
568 investigate. What this framework adds to that investigation is an order for
569 the questions: declare the family of states and the fidelity, fix the encoding
570 region and read its causal past off the architecture, and only then ask what
571 the system reconstructs. A claim that survives those three steps is a claim

about the machine; one that does not survive them was a claim about its
output.

7 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them, and it does so from the side rather than along the spine: no edge below derives coherence, compatibility or reconstruction from a heat bound, and the resource condition these edges terminate in is necessary without being restrictive for any macroscopic candidate (Section 3.2). What this half contributes is a ledger in which a resource claim is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

7.1 Probability current supplies a dissipation bound

For a positive normalized phase density ρ , diffusion $D > 0$ and current $J = v\rho - D\partial_\theta\rho$, define $\sigma_J = \int_{-\pi}^{\pi} J^2/(D\rho) d\theta$. Continuous density and current give the sharp bound

$$\left(\int_{-\pi}^{\pi} J d\theta\right)^2 \leq D\sigma_J. \quad (12)$$

591 A uniform density under constant drive ω attains $\sigma_J = \omega^2/D$. In contrast,
 592 the identical-frequency gradient model has positive self-consistent stationary
 593 order above $K = 2D$ with $J = 0$ and $\sigma_J = 0$. Thus neither positive order nor
 594 its time average forces positive dissipation: a stationary ordered density is
 595 already a counterexample. A cortical maintenance-power bound requires an
 596 identified nonequilibrium drive, its probability current and a thermal con-
 597 version; phase order alone does not supply them. The current functional
 598 is distinct from the substrate squared-drift objective used in the plasticity
 599 controls. It also bounds the rate of the cosine moment $m = \int \cos \theta \rho(\theta) d\theta$:
 600 $\dot{m}^2 \leq D\sigma_J$, when the classical continuity equation and sufficient regular-
 601 ity justify differentiating this moment. This constrains density change. It
 602 does not bound an externally chosen $\dot{K}$: the uniform density is an exact time-
 603 dependent solution of zero current cost for every coupling schedule, including
 604 positive schedules with any prescribed derivative.

605 7.2 Feedback changes the information constraint

606 For passive memory X_t driven by an environmental signal S_t , the predictive
 607 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (13)$$

608 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 609 et al., 2012). Zero retained information also satisfies it, so a task and a
 610 learning rule require additional specification. Relating this bound to Lan-

611 dauer heat requires a common thermal scale and budget, and memory waste
 612 bounded by erased entropy.

613 An acting agent changes the signal it will receive. Specify policy, world,
 614 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{14}$$

615 The second conditional term vanishes in the passive model. With feedback,
 616 the signed difference can be negative as action creates correlations with the
 617 future. A causal reading requires an environmental state sufficient to describe
 618 the relevant dynamics.

619 For a controlled substep, the active thermodynamic bound instead uses
 620 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 621 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
 622 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
 623 Ito and Sagawa, 2013). Every forward path of positive probability must have
 624 a positive-probability reverse for this finite identity. A missing reverse gives
 625 infinite divergence. The supplement develops the support conditions and
 626 composes actuation with memory updates on their shared intermediate law.

627 **7.3 Learning and repeated operation need explicit re-** 628 **sources**

629 The finite examples distinguish selecting a feasible policy, relaxing a register
630 toward an encoded reward, and learning an unknown task from observa-
631 tions. The observational learner acquires correlation with an environmental
632 parameter while its action marginal stays uniform. A continuing model exe-
633 cutes actuation, observation-driven learning and reset in sequence, preserving
634 above-chance performance at a positive work cost per cycle.

635 A phase-derived sensor makes the informational distinction concrete: an
636 informative readout and a blind control have the same phase order but dif-
637 ferent learning performance. Attaching the learner’s register to the actuator
638 of Section 3.2 ties reward, installed kernel and work to its own executed law.
639 The learning witness rises above the necessary installed-energy bound while
640 its matched control remains on the exclusion line. These finite constructions,
641 detailed in Supplemental Material, leave the evolving feedback between phase
642 dynamics, installation and sensing to be specified.

643 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
644 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (15)$$

645 where expected cumulative work stays within the supplied allowance B at
646 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.4 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one:

666 strictly positive forward and reverse elementary paths, local detailed balance
 667 at one thermal scale, and entropy differences and heats of order $k_B T$, the
 668 witnesses being two-bit systems with interaction energies of that size. Cortex
 669 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
 670 temperature over one region and one phase-diffusion time, and a budget B
 671 of that size stands against an entropy term of order $k_B T$, so Eq. (16) is
 672 satisfied by any macroscopic candidate and excludes none of them. Reading
 673 the register instead at the scale where its premises are physical, such as a
 674 single channel, relocates the difficulty to E45Active, which would then have to
 675 carry one molecular step’s budget to the convergence of a continuum kernel.
 676 The active branch is an alternative route to the same node rather than a
 677 stronger one, and no result of Section 3.2 passes through it: the installed-
 678 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

679 Both branches also take a thermodynamic cover carrying exact over-
680 lap compatibility and a reflexive boundary specifying encoding and readout.
681 They conclude **UnifiedSelf**: the glued section is the unique fixed point of

self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named in one place and machine-checked together, and not that coherence follows from installed energy or self-representation from coherence.

Joint witnesses combine a register, boundary, refining mesh and small spatial cover; active witnesses add controlled thermal updates and a priced installation. They establish satisfiability under their stated identifications. Mesh convergence follows from refinement data, and the witnesses establish no common cortical mechanism. Supplemental Material maps the claims to their Lean declarations and records the counterexamples. The development declares no additional axioms beyond its libraries' foundational axioms.

703 8 What the observations establish

704 The composition holds under its premises. Whether observation can estab-
705 lish those premises is a separate question, and the three requirements it
706 connects concern different objects: phase dynamics, local descriptions and
707 internal encodings. Their observables must retain these distinctions. The
708 following controls show which mechanisms remain indistinguishable, which
709 temporal approximations fail, which states one observable cannot separate,
710 and what independent content measurements must establish. Each is a finite
711 numerical regime on a declared model, with full parameters and controls in
712 Supplemental Material, and none of them calibrates a cortical quantity.

713 8.1 Stationary distributions leave the mechanism un- 714 determined

715 Equation (2) gives a concentration–coherence relation with no fitted shape
716 parameter. Estimate r from circular resultant length and a from density
717 shape: inverting R from the measured r would make agreement automatic.
718 Even independently estimated agreement identifies a statistical family shared
719 by different mechanisms.

720 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
721 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
722 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
723 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

724 and diffusion D_0 have that same density. Their stationary distributions co-
 725 incide window by window despite different causes.

726 Resolved dynamics supply additional information. In the ideal model with
 727 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 728 increment means constrain drift. Separating endogenous coupling from a
 729 matched common drive additionally requires a measured drive or controlled
 730 perturbation; identifying electromagnetic coupling requires evidence specific
 731 to that interaction. Supplemental Material gives the equivalence construction
 732 and its numerical check.

733 8.2 Finite ramps change the observable onset

734 The square-root law in Eq. (3) describes stationary solutions. A changing
 735 coupling requires time for the phase distribution to approach those solutions.
 736 In a deterministic mean-field control that crosses the threshold symmetrically,
 737 none of the scanned rates meets the declared tracking tolerance: slowing
 738 the crossing does not remove the terminal residual under that initialization.
 739 A slow control variable therefore supplies no general guarantee of branch
 740 tracking at onset. Supplemental Material gives the residual, rate range and
 741 admissible rates away from threshold.

742 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 743 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 744 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 745 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The

746 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 747 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 748 onset exponents differ from those of the exact stationary branch evaluated
 749 on the same grids. At the fastest speed the ensemble mean never reaches the
 750 fit window, so its fitted slope does not estimate stationary onset.

751 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 752 makes the apparent exponent depend on ramp speed and fitting window.
 753 The specified deviation-based critical ramp speed is not bracketed by this
 754 sweep. With fluctuating coupling, the relevant stationary comparison also
 755 depends on rate: the mean coupling approximates order in the tested fast
 756 regime, while averaging stationary order over coupling works for the tested
 757 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 758 neither approximation.

759 **The prediction.** The delay scaling is what this section supports as a pre-
 760 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 761 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494],
 762 which does not contain 1/2. What that shortfall measures is the escape cri-
 763 terion and not the scaling. Applied to a deterministic mean-field run seeded
 764 at the fluctuation floor, the same estimator returns 0.425 over the same legs
 765 at the same escape criterion, inside the interval on the measurement. That
 766 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
 767 saturation between the floor it starts from and the level it is stopped at, and

768 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
769 where the answer is known, which is what licenses reading the measurement
770 as the law rather than as a competing exponent.

771 Stated as a law rather than as a number: onset delay scales as the square
772 root of the rate at which coupling crosses threshold. Testing it requires
773 only that $\dot{K}$ be monotonic in a manipulated rate—not its value, nor the
774 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (17).
775 It is therefore a within-subject design varying the rate of emergence from
776 anaesthesia alone, each participant serving as their own control. Such a
777 design must avoid the hazard this sweep exhibits: a fixed absolute escape
778 criterion meets a size-dependent critical fluctuation floor, so the criterion
779 has to be referred to each recording’s own pre-threshold fluctuation scale
780 rather than to a shared number. The diffusion D is the remaining input,
781 and Section 8.1 notes that it is estimable from resolved phase traces through
782 the rate $2D$ at which quadratic variation accumulates, which is a different
783 quantity from the concentration the propofol exercise could not range over.

784 Additional controls address integration bias, population size and spatial
785 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
786 $[1.958, 2.009]$ contains 2.

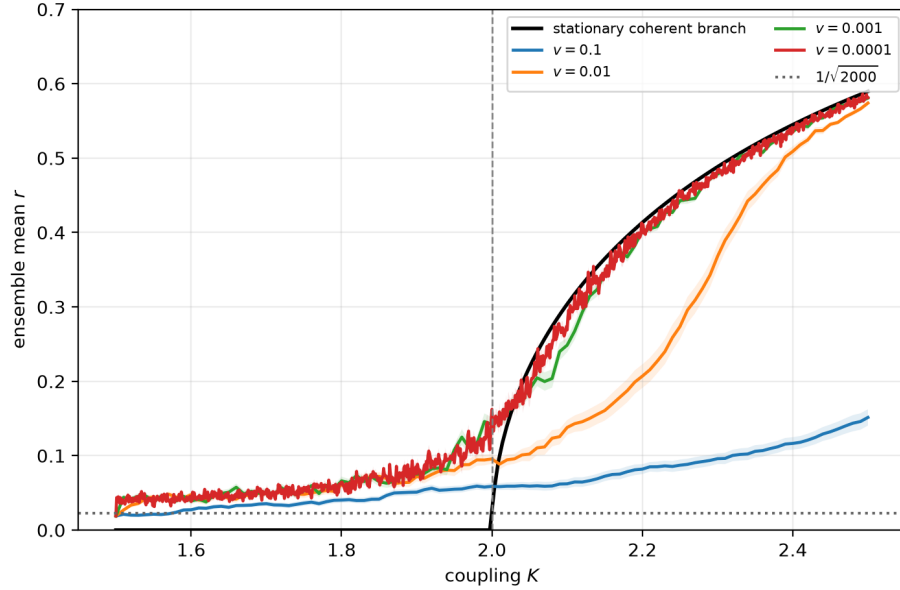


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

787 8.3 A winding state leaves the order parameter am- 788 biguous

789 The order parameter r is indifferent to where a phase sits in space. A field
790 winding once across a periodic sheet therefore has $r = 0$ while every patch
791 of it stays locked. Such a field is not an artificial construction here. On a
792 torus a winding is stationary for any isotropic symmetric kernel at identical
793 frequencies: the coupling term is odd in the separation against a kernel even
794 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves

795 this for any finite periodic array, this sheet included. A winding satisfies the
796 positivity and identical-frequency hypotheses. It violates only the quarter-
797 turn initial spread that the convergence result additionally requires before
798 placing the limit at the uniform state.

799 On the sheet of Section 3's spatial control, a winding survives below
800 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
801 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
802 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
803 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
804 patch-local order 0.9015 and no winding defect. At fixed K the observable
805 therefore takes two values spanning nearly its whole range. Coupling does
806 not determine the resultant, and a low resultant does not by itself distinguish
807 weak coupling from a winding field.

808 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
809 trol carries, the boundary moves to 0.1421 mm. The same comparison at
810 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
811 Heterogeneity shortens the range that sustains a wave without removing it.

812 Which windings persist has a criterion behind it. Linearizing the phase
813 model about a winding returns a coupling of the same translation-invariant
814 form, each separation reweighted by the cosine of the phase the winding
815 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
816 again even in the separation, so the same characters diagonalize it and ev-
817 ery mode's rate is one of its eigenvalues. Where the winding advances by

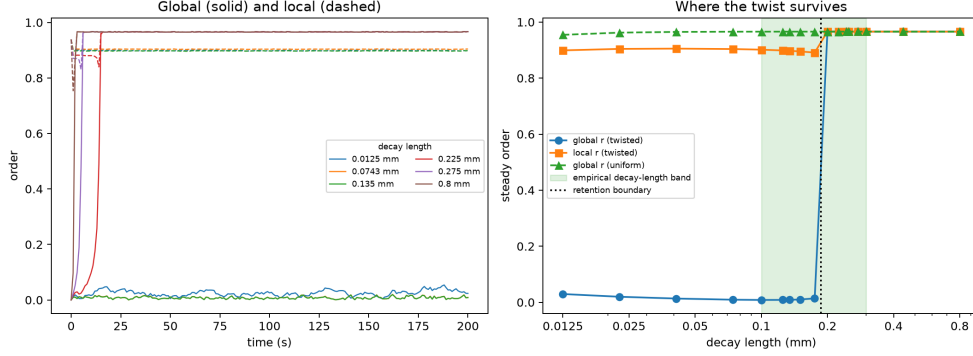


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

818 at most a quarter turn across every separation the coupling reaches, the
 819 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 820 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 821 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 822 longer separations is what carries a phase advance past a quarter turn, and
 823 that is the mechanism the decay-length sweep varies. The criterion bounds
 824 rates rather than trajectories: asymptotic stability is not proved, so the
 825 states that persist occupy a narrower range than it admits, and the bound-
 826 ary quoted above is measured and not derived from it. The same kernel is the
 827 potential's curvature, so a winding at $r = 0$ is not the global minimum the

828 convergence result places its limit at, and one carrying a growing mode is not
 829 a local minimum either: that result's quarter-turn hypothesis must exclude
 830 a winding rather than merely decline to cover it (`char_not_potential_min`).

831 An amplitude layer decides which of these states exist, and leaves what
 832 the resultant reports unchanged. Every oscillator of the phase model sits on
 833 the unit circle, so a field with an amplitude node has no representation in
 834 those dynamics. The real Stuart–Landau extension supplies one on the same
 835 coupling, and leaves intact the gradient structure the convergence results rest
 836 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 837 phase equation is the Kuramoto equation with each coupling weighted by a
 838 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 839 this one with the amplitude ratios set to one, rather than a separate object.

840 What the added degree of freedom decides is which windings exist at all.
 841 On a periodic array of any dimension, this sheet included, a winding requires
 842 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 843 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 844 phase the winding advances across d (`exists_charState_iff`; on the ring
 845 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 846 coupling mass to longer separations lowers it further. A broad enough kernel
 847 therefore removes the winding outright. The sweep above reaches that same
 848 boundary by a different route: the phase model retains every winding at
 849 every coupling, and loses this one under integration instead. The two layers
 850 bound the same regime from opposite sides. A phase recording does not

851 separate a state the amplitude layer excludes from one the dynamics fails to
852 sustain.

853 Three limits bound the layer for a reader of the cortical literature. These
854 states are windings and not spirals, and on the sheet that is a theorem rather
855 than a shortage of dimensions. Equal modulus at every site puts the phase
856 circulation at exactly zero on every plaquette, and on a periodic sheet the
857 circulations of *any* field sum to zero, so a lone defect is not a state of this
858 geometry at all. Relaxational dynamics has no propagation speed, which is
859 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
860 form radius, which a measured envelope is not; this work identifies the two
861 nowhere. Supplemental Material gives the sweep, its uniform control, the
862 amplitude layer, the standing wave it admits inside the band, the potential's
863 curvature at a winding and the defect theorems, together with the remaining
864 limits of both.

865 A vanishing resultant is the absence of global agreement and not of agree-
866 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
867 ancy bound of $2\sqrt{2} L |\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
868 which falls to zero as the same winding is spread over more sites (`chord_le_`
869 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
870 Encoders shared on overlaps therefore still bound content disagreement in-
871 side a patch of a state that no global observable calls coherent. The winding
872 number itself stays unconnected to content. It is, however, a property of the
873 state rather than of the cover read on it. On a ring the whole turns dropped

874 by the winding's affine local lifts are zero at every site but the wrap, where
875 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
876 every global real-valued phase lift, so one state at one coupling has resultant
877 exactly zero and degree q (`winding.degree obstructs`). Reweighting the
878 sites moves the first quantity and redrawing the loop's intermediate transi-
879 tions leaves the second where it was. The scope is the ring's: this is a loop's
880 winding and not a spiral, the transitions are those of the affine lifts rather
881 than of an unwrapped measurement, and the integer is identified with no
882 cortical observable.

883 This qualifies what a phase trace can establish. An onset criterion stated
884 in r alone registers a change of basin as readily as a change of coupling.
885 The protocol of Section 9.3 therefore needs a spatial observable beside the
886 resultant.

887 **8.4 Coherent adaptation can lose environmental align-** 888 **ment**

889 An adaptive coupling rule might supply representational structure, but its
890 objective must reward the relevant agreement. For symmetric coupling, pair-
891 wise sine terms cancel in the population mean drift. Minimizing squared drift
892 at that fixed mean therefore rewards equalizing drift across nodes. In the
893 frequency-cluster control, this favors coupling between unlike-frequency clus-
894 ters, whereas the diagnostic environmental template rewards coupling within

895 clusters.

896 The formalization proves descent of the squared-drift objective for a spec-
897 ified gradient flow at fixed phases. Its functional differs from thermodynamic
898 entropy production: at a coherent stationary density the probability current
899 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
900 rial gives the operator, its gradient and the conditions of the descent result.

901 The finite control co-evolves noisy phases and symmetric, nonnegative
902 couplings at fixed total resource. Its template is a diagnostic covariance
903 of latent signals, never a training input. Across three illustrative seeds,
904 second-half order remains between 0.9167 and 0.9222, while the within-
905 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
906 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
907 trols are reported in Supplemental Material.

908 The immediate post-update objective reduction disappears when aver-
909 aged over complete update intervals. A prespecified grid of 6 rate–cadence
910 combinations contains no case meeting the 5% sustained-reduction criterion
911 while retaining coherence and losing alignment. The finding is coherent adap-
912 tation away from the specified structure, with no demonstrated sustained cost
913 reduction.

914 Kernel alignment and overlap compatibility are different observables.
915 This control addresses acquisition of its designated template under its par-
916 ticular objective. Learning models can explicitly include data in an objective
917 (Corberi et al., 2026); an account of agreement about the cup must specify

918 the content its learning rule is meant to preserve.

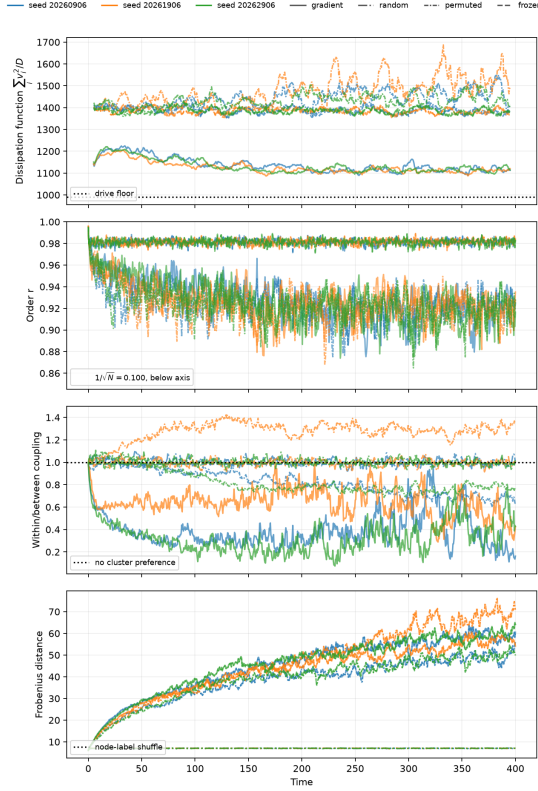


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

919 8.5 Content agreement needs independent decoders

920 A neural test needs independently identified domains, local descriptions and
921 restriction maps. Two decoders should estimate a shared content on over-
922 lapping domains before their agreement is examined. Discrepancy requires a
923 null under decoding noise and controls using deliberately compatible and in-
924 compatible examples. An implementation on synthetic data separates these
925 cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
926 the compatible construction scores 0.476 on an interval containing one half.
927 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
928 bias can manufacture disagreement. Per-patch decoding diagnostics detect
929 these failures; the compatibility statistic alone does not.

930 Phase agreement cannot replace that measurement. On constructed con-
931 tent, phase carries a share 1.000 of the content entropy when content is built
932 from it and 0.000 when it is not. A phase-derived statistic reports com-
933 patibility on phase-locked patches whose local descriptions disagree. The
934 implemented content statistic detects their incompatibility. These controls
935 validate a measurement on known constructions; a neural content model and
936 decoder remain to be established.

937 Choosing the cover also chooses a candidate subject: trivial descriptions
938 can agree while carrying little content, and a partition can conceal disagree-
939 ment. The useful choice must therefore be constrained by what neural ac-
940 tivity encodes. The exact composition assumes compatibility of its supplied
941 cover. The content mechanisms above identify additional hypotheses under

942 which agreement can arise; they leave the empirical choice of content and
943 the connection to that cover explicit.

944 **8.6 Estimator calibration on exploratory EEG**

945 The concentration range and the dependence structure decide whether the
946 concentration-coherence relation can be tested at all. Under the specified
947 log-density estimator and bias-floor criterion, discrimination at the observed
948 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
949 designs, rising to 31000 when sites share a mean in groups of 10. These
950 counts describe that estimator and dependence model; the intended recording
951 requires its own calibration. Supplemental Material reports the bin-count
952 controls and the full dependence-aware design.

953 That ceiling comes from exercising the estimators on public propofol-
954 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-
955 dows. Bipolar and common-reference montages produce different residuals.
956 Bipolar-montage scalp EEG sits in the linear Bessel regime ($a_{\max} = 0.542$,
957 deviation from $a/2 < 1\%$). The exercise confirms the estimator is well-
958 behaved in this regime but does not test the nonlinear prediction. The
959 bipolar results are compatible with the Bessel relation and with any mono-
960 tone alternative in this regime. Temporal pooling changes the estimand, and
961 sedation recordings leave the awakening hypothesis untested.

962 An informative dataset therefore needs sufficient concentration range or
963 precision, calibrated estimators and an observation model for dependence,

964 source mixing and reference choice. Testing the nonlinear prediction requires
965 intracranial recordings (ECoG, sEEG or LFP) where local synchrony reaches
966 $a > 1$, or narrowband alpha-spindle burst analysis, at the sample sizes es-
967 tablished above ($\geq 31,000$ pooled phase samples or ≥ 1000 independent
968 sites). Public mouse calcium imaging provides thousands of spatially located
969 neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its
970 7.65 Hz frame rate is below the phase-estimation band used here. Site count,
971 state coverage and bandwidth are separate requirements.

972 **9 The cortical hypothesis and an awakening**

973 **test**

974 The cortical proposal identifies a physical source of coupling and asks whether
975 its changes help explain recovery. The tests in Section 8 determine what
976 observations must distinguish before a phase trace can support this identifi-
977 cation.

978 **9.1 Physical identification and calibration**

979 The cortical hypothesis identifies part of effective coupling with endogenous
980 extracellular electromagnetic interactions. Fields can modulate spike timing
981 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
982 propose coordination of distributed neural information (Pinotsis & Miller,

2022; McFadden, 2020). Phase organization under propofol motivates study-
ing this regime (Bastos et al., 2021), but does not measure coupling.

The formal field predicate requires a jointly continuous kernel, a normal-
ized substrate, coupling equal to the kernel average, and no single site pro-
viding more than half that strength. Toy constructions satisfy it. A cortical
realization must specify the scale at which the kernel and its reduction ap-
ply: calcium imaging reports different state-dependent modularity patterns
at single-cell and mesoscale resolution (Kiyooka et al., 2026).

Cortical columns are the obvious candidate unit, and they answer one
of the two demands the formalism makes. As the oscillator index they are
a serious proposal: they are delineated by response properties and anatomy
rather than by the agreement they would be used to establish (Mountcastle,
1997), and a phase can be estimated from a column-sized territory at the
electrode spacings used for cortical wave measurements (Townsend et al.,
2015). As the content cover they are a poor one, for three separate reasons.
A parcellation into non-overlapping columns is a disjoint cover, which satisfies
the compatibility obligation without constraining anything (Section 4). The
cover condition additionally requires the patches to exhaust the substrate,
while much of association cortex has no accepted columnar parcellation. And
the column’s standing as a canonical cortical unit is contested (Horton &
Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
supplies the oscillator index and leaves the content cover to be identified on
separate evidence, which is the bridge assumption E78 records.

1006 The spatial reduction is also testable. A row-sum aggregation at zero
 1007 diagonal tracks the uniform-coupling control over the tested spatial decay
 1008 range, while frequency heterogeneity displaces the operational threshold. The
 1009 fit window biases both estimates relative to $2D$; agreement with the control
 1010 is therefore the supported conclusion. Supplemental Material reports the
 1011 reduction, its numerical error and the distinct assumptions needed for the
 1012 kernel-average identification in Eq. (5).

1013 The field commitment requires physical evidence beyond a choice of math-
 1014 ematical representation, and the available support comparisons do not supply
 1015 it. A finite matrix spread over positive-mass cells preserves its total cou-
 1016 pling and phase correlation exactly, while point-supported vanishing under
 1017 an atomless measure and a disadvantage from excluding a better-correlated
 1018 pair concern specified support constraints; Section 6 reads the same compar-
 1019 isons against a digital candidate. Neither supplies energy prices.

1020 A spike-time shift alone supplies no coupling rate. The dimensionless
 1021 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 1022 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (17)$$

1023 This rate describes how induced timing shifts accumulate into drift. Nei-
 1024 ther it nor the installation factor κ follows from a heat bound. Testing the
 1025 physical threshold requires independent estimates of the interaction kernel,

1026 its aggregation and diffusion in the same cortical regime.

1027 **9.2 Extracellular geometry and recovery**

1028 The proposed control variable is extracellular geometry. Extracellular vol-
 1029 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 1030 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 1031 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 1032 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 1033 an effective-medium description predicts that a larger extracellular fraction
 1034 raises conductivity and reduces field amplitude.

1035 The coupling-gated hypothesis consequently predicts that extracellular
 1036 contraction on awakening raises E and, holding the other factors in Eq. (17)
 1037 fixed, raises effective coupling. This argument supplies a direction; the
 1038 geometry–coupling relation, its magnitude and its role in human recovery
 1039 require measurement.

1040 For a smooth crossing at nonzero speed, evaluating the stationary branch
 1041 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (18)$$

1042 This follows from Eq. (3) at fixed D and concerns the stationary branch
 1043 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 1044 dently justified tracking approximation. The crossing controls in Section 8.2

1045 leave that approximation unsupported at onset, even though a separation of
1046 geometry and phase time scales can support it away from threshold. The
1047 awakening hypothesis therefore needs a calibrated dynamical prediction for
1048 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
1049 does not supply a recovery latency.

1050 **9.3 A protocol for the awakening experiment**

1051 The resistance of the central nervous system to state transitions, known
1052 as neural inertia, manifests as a behavioural barrier and hysteresis between
1053 anaesthetic induction and emergence (Friedman et al., 2010). The recovery
1054 of responsiveness is not linear, but rather mediated by stochastic transitions
1055 through a network of metastable activity states (Hudson et al., 2014; Proekt
1056 & Hudson, 2018). While the dynamic sweep of a bifurcation parameter pro-
1057 vides a distinct temporal scaling, the $v^{1/2}$ exponent derived in Section 8.2
1058 is a generic feature of delayed supercritical bifurcations (Baer et al., 1989)
1059 and cannot by itself discriminate a field mechanism from a synaptic one.
1060 The framework’s discriminating content is instead the spatial onset pattern:
1061 coherence must nucleate in regions of highest κ (mode density times field
1062 strength) and correlate locally with field amplitude, rather than appearing
1063 first at the structural hub nodes a synaptic-only model predicts.

1064 Testing this requires a within-subject protocol manipulating emergence
1065 rate, with high-density intracranial recordings (e.g., ECoG or Neuropixels)
1066 to resolve the spatial signature:

- 1067 **1. Manipulate emergence rate.** Vary the rate v of anaesthetic (e.g.,
1068 propofol) clearance via infusion rate across multiple sessions within the
1069 same subject.

- 1070 **2. Measure behavioural recovery.** Record time-to-response to a stan-
1071 dard stimulus as the independent behavioural endpoint, rather than
1072 estimating ΔK directly.

- 1073 **3. Resolve spatial onset.** Identify the precise anatomical locations
1074 where phase coherence first reaches threshold. The competing null
1075 is that onset occurs at the connectome’s hub nodes or uniformly; the
1076 field hypothesis predicts onset in high- κ regions where extracellular
1077 geometry best supports field coupling.

- 1078 **4. Match physical controls.** Confirm that the observed coherence-
1079 to-amplitude relation holds under the specific local diffusion D and
1080 observation noise, estimable from the quadratic variation of resolved
1081 phase traces.

- 1082 Failure of the onset location to track regional κ , or better prediction of
1083 spatial onset by structural network centrality, falsifies the field coupling mech-
1084 anism. A favorable outcome still leaves the content of coordinated activity
1085 to be explained.

1086 10 Discussion

1087 Two things have been established here, and they are of different kinds. The
1088 constraints of Section 2 hold outright: synchrony does not deliver compatible
1089 content, and where shared encoders let it bound content at all, the bound
1090 that holds uniformly over pairs does so only inside patches too small to
1091 span the territories at issue, and distance costs either a connected cover or
1092 a locked configuration with a declared spectral gap. The resultant does not
1093 determine the state; a unique fixed point is not a reconstruction; a region that
1094 cannot resolve a family cannot encode it; and agreement about a change takes
1095 communication. Any account that reads unity off phase coherence has to
1096 meet them, this one included. The account set out in Section 1 is conditional
1097 at every step by contrast: each connection in it is a hypothesis supplied
1098 rather than a result derived, and Table 1 names all eight. Its four non-
1099 standard ingredients, named in Section 1, are the installed-energy condition,
1100 the patch-local form of the content bound, the criterion separating a winding
1101 field from weak coupling, and the machine-checked composition with its joint
1102 witness. What follows takes the conditions in turn, as commitments the
1103 proposal incurs, as measurements that would bear on them, and as questions
1104 the framework leaves open.

1105 **10.1 Unity and a minimal self as interpretive motiva-** 1106 **tion**

1107 We identify the state glued from compatible local descriptions as a formal
1108 correlate of conscious unity, and its accurate internal reconstruction as a
1109 correlate of a minimal self. This identification is the framework’s explicit
1110 interpretive motivation: it explains why the mathematics might matter. The
1111 motivations are that different contents belong to one experienced situation
1112 and that the situation is experienced from a perspective. The framework’s
1113 empirical content, however, is the coupling-gated onset prediction and its spa-
1114 tial signature. The conditional results identify relations a realization would
1115 need: sufficient installed resources for the specified phase model, agreement
1116 about shared content, and an encoding that supports reconstruction at the
1117 required fidelity. Discharging E12–E89 for cortex would make the composi-
1118 tion a statement about cortex: that a cortical field above threshold relaxes
1119 to a cover whose compatible local descriptions glue, and that the glued state
1120 is the unique fixed point of a reconstruction map realized there. It would
1121 leave the two identifications above exactly where they are, because neither is
1122 among the eight and neither follows from their conjunction; they are the mo-
1123 tivation this subsection names, and without an independent measurement for
1124 experience, they do not form a testable proposal. Section 10.2 lists the em-
1125 pirical observations that would count against the framework’s mathematical
1126 claims.

1127 Compatibility allows different descriptions to contribute different infor-
1128 mation. It also permits uncertainty and conflicting judgments: apparent cup
1129 position and judged cup position can be distinct quantities in a consistent
1130 description of their discrepancy. Which quantities are shared must be de-
1131 termined independently. Choosing them solely to secure agreement would
1132 deprive the proposal of empirical force.

1133 The minimal-self proposal concerns the perceiver’s position and condition
1134 within the situation, such as the cup’s location relative to a body that can
1135 reach it. The encoding could be implicit and need carry no autobiograph-
1136 ical narrative. Its proposed necessity is philosophical: even an unreflective
1137 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1138 constructability alone supplies no functional role. Altering the encoded hand
1139 or viewpoint while holding the external scene fixed should change estimates
1140 of the perceiver’s relation to the cup as the readout predicts. Whether that
1141 represented perspective is experienced remains a further question.

1142 **10.2 What a test of the full proposal must establish**

1143 Both uniqueness results concern specified objects. They choose neither the
1144 cover nor the encoding, and do not select between candidate subjects. In-
1145 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1146 here selection requires independent evidence about neural organization. A
1147 constant readout can contract and fix a predetermined state while discard-
1148 ing its input. The family criterion in Section 5.2 tests the sensitivity that

1149 this construction lacks, provided the relevant states and fidelity are declared
1150 independently.

1151 A test must separately measure coupling relative to diffusion, content
1152 compatibility and self-reconstruction, alongside an independent endpoint for
1153 experience. Meeting all three without a unified episode would challenge suffi-
1154 ciency. A unified episode with incompatible descriptions would challenge the
1155 unity identification; one lacking the specified self-encoding would challenge
1156 the reflexive requirement. Awakening, dreaming and dissociative states offer
1157 possible settings, but absence of report cannot by itself define absence of
1158 experience.

1159 The next physical requirement is a calibrated geometry–coupling relation
1160 and resolved phase dynamics. The installation bound contributes a nec-
1161 essary resource condition where its hardware representation applies; cortical
1162 mode prices and field-mediated coupling require measurement. For the repre-
1163 sentational connection, an informative experiment asks whether a controlled
1164 change in coordination alters independently decoded content agreement, with
1165 controls for sensory information and decoder quality. The constructed-data
1166 estimator establishes feasibility for specified contents. Neural local descrip-
1167 tions, shared overlap encoders and the self-encoding have yet to be identi-
1168 fied. The population-size dependence of the phase-to-content bound limits
1169 its quantitative use in the form Section 4.2 reads off the sheet, and each
1170 way around it asks the measurement for something further: a located excep-
1171 tional set, before a count of disagreeing pairs bears on a named overlap; a

cover whose nerve is connected and whose patch diameters are measured; or
a frequency-locked population over a coupling graph whose connectivity is
measured structurally rather than estimated from the activity the bound is
about.

The separation results explain why these measurements must be independent. A stationary distribution leaves several recovery mechanisms indistinguishable. Equal phase order can accompany incompatible descriptions or different task information. The tested plasticity rule preserves coherence while losing its designated template. Each finding concerns the specified model and observable; their role is to delimit the evidence needed for the proposed connections.

10.3 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. That permission is wider than the unconditional results of Section 2 alone. The conditional chain is substrate-neutral in the same way: its field condition asks for a jointly continuous kernel on a normalized substrate, a coupling equal to that kernel’s double average, a noise that is the field’s own, and

1194 no single site carrying more than half the coupling—and none of these dis-
1195 tinguishes an electromagnetic kernel from any other continuous mean-field
1196 kernel. The site condition is the only one with physical flavour, and it holds
1197 vacuously wherever the substrate has no atoms (`no_site_dominates`), which
1198 is the continuum case the electromagnetic hypothesis is about. The chain
1199 therefore transfers to extracellular ion diffusion, to gap junctions modelled
1200 as a density, or to a mechanical or optical medium, modulo the calibration of
1201 Section 9.1. What carries the electromagnetic identification is that calibra-
1202 tion together with the extracellular-geometry argument of Section 9, which is
1203 the one place a field equation enters. The formalization checks implications
1204 under their hypotheses, and its witnesses establish realizability on declared
1205 models. Two things stand outside that result and are not supplied by it: the
1206 eight hypotheses are discharged for toy components rather than for cortex,
1207 and the identification of the glued state with unity and of its reconstruction
1208 with a minimal self is a further commitment that the eight do not contain.

1209 The explanatory question is whether the specified relations occur in neural
1210 activity and matter for experience. In the cup example, this means establish-
1211 ing how coordination helps maintain one represented situation and how the
1212 perceiver’s place within it contributes to the episode. The proposal identi-
1213 fies candidate correlates and tests of that identification. Why those relations
1214 would be accompanied by experience remains unexplained.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from

Table 2 (continued)

Symbol	Defined	Meaning
$k_B T, B$	Eq. (16)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (17)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
$\mathcal{E}$	Eq. (15)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (17)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

1216 Data and code availability

1217 The Lean development, simulation scripts, tests and saved numerical sum-
 1218 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`

1219 `cortical_coherence`. Publication simulation values are generated from
1220 saved summaries without rerunning production sweeps. The exploratory
1221 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025) (sub-
1222 jects 1010, 1016, 1022, 1033, 1045, 1056, resting awake and sedation runs
1223 1–2); preprocessing and selection are described in Supplemental Material
1224 and implemented in `simulations/empirical_collapse.py`. The repository
1225 contains the reproducible analysis and does not redistribute the raw record-
1226 ings.

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1230 **Use of AI tools**

1231 AI assistance to this work was substantial and is not separable by task. Sev-
1232 eral coding assistants and chat interfaces were used in an interleaved way
1233 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1234 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1235 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1236 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1237 accessed with the Hermes CLI. Between them they contributed to the Lean

development, the simulation and analysis code, and the drafting and revision of both documents. Because the tools were used in combination and in alternation, no section, proof, figure or result is attributable to any one of them, and none is claimed to be. The author is responsible for all content, including everything a tool produced. Machine checking establishes the stated Lean results under their hypotheses; it does not validate biological interpretations or replace scientific review of generated code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.

- 1257 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
1258 *Reviews Neuroscience*, 11(2), 127-138.
- 1259 Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness:
1260 A case for biological computationalism. *Neuroscience & Biobehavioral Re-*
1261 *views*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- 1262 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
1263 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
1264 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.
- 1265 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
1266 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 1267 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
1268 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
1269 doi:10.1017/S0140525X26106906.
- 1270 Gurnee, W., et al. (2026). Verbalizable Representations Form a
1271 Global Workspace in Language Models. *Transformer Circuits Thread*.
1272 arXiv:2607.15495.
- 1273 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
1274 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
1275 40, 24-50.
- 1276 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
1277 parts of the free energy principle. *Entropy*, 23(3), 293.

1278 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
1279 peror's New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

1280 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1281 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

1282 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1283 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
1284 *Progress in Neurobiology*, 226, 102465.

1285 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1286 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1287 9(8), 45-60.

1288 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1289 testable hypothesis about the characteristics of conscious as opposed to
1290 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

1291 McFadden, J. (2020). Integrating information in the brain's EM field: the
1292 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1293 niaa016.

1294 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1295 guide neocortical network activity. *Neuron*, 67(1), 129-143.

1296 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1297 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1298 95-103.

- 1299 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1300 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 1301 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1302 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 1303 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1304 *logical Reviews*, 98(1), 239-389.
- 1305 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
1306 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 1307 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
1308 without a function. *Philosophical Transactions of the Royal Society B*,
1309 360(1456), 837-862. doi:10.1098/rstb.2005.1623.
- 1310 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
1311 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.
- 1312 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1313 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1314 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1315 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
1316 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
1317 ity of complexity measures to capture dreaming during propofol sedation.
1318 *Scientific Reports*, 15, 32746.

- 1319 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiagarajan, M., ... &
1320 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1321 brain. *Science*, 342(6156), 373-377.
- 1322 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
1323 *Physiological Reviews*, 88(4), 1277-1340.
- 1324 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
1325 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.
- 1326 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1327 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1328 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- 1329 Landauer, R. (1961). Irreversibility and heat generation in the computing
1330 process. *IBM Journal of Research and Development*, 5(3), 183-191.
- 1331 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1332 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 1333 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1334 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- 1335 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1336 of prediction. *Physical Review Letters*, 109(12), 120604.
- 1337 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1338 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

- 1339 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1340 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1341 031015.
- 1342 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1343 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1344 PhysRevLett.111.180603.
- 1345 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1346 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 1347 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1348 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1349 181.
- 1350 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1351 archive.pitt.edu/115.
- 1352 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1353 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1354 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1355 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1356 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1357 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1358 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork

1359 bifurcations with additive noise. *Probability Theory and Related Fields*,
1360 122, 341–388. doi:10.1007/s004400100174.

1361 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1362 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1363 doi:10.1088/1367-2630/13/11/113036.

1364 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1365 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
1366 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
1367 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

1368 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
1369 terns underlie complex brain dynamics and are related to cognitive pro-
1370 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
1371 023-01626-5.

1372 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
1373 and dissipation with partially observed fluctuating signals. *Physical Review*
1374 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

1375 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
1376 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
1377 cell resolution functional networks during unconsciousness are segre-
1378 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
1379 doi:10.1016/j.celrep.2025.116902.

- 1380 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
1381 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
1382 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.
- 1383 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
1384 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
1385 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.
- 1386 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1387 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.
- 1388 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.
- 1389 Baer, S. M., Erneux, T., & Rinzel, J. (1989). The slow passage through a
1390 Hopf bifurcation: delay, memory effects, and resonance. *SIAM Journal on*
1391 *Applied Mathematics*, 49(1), 55–71. doi:10.1137/0149003.
- 1392 Friedman, E. B., Sun, Y., Moore, J. T., Hung, H.-T., Meng, Q. C., Perera, P.,
1393 Joiner, W. J., Thomas, S. A., Eckenhoff, R. G., Sehgal, A., & Kelz, M. B.
1394 (2010). A conserved behavioral state barrier impedes transitions between
1395 anesthetic-induced unconsciousness and wakefulness: evidence for neural
1396 inertia. *PLoS ONE*, 5(7), e11903. doi:10.1371/journal.pone.0011903.
- 1397 Hudson, A. E., Calderon, D. P., Pfaff, D. W., & Proekt, A. (2014). Recovery
1398 of consciousness is mediated by a network of discrete metastable activity
1399 states. *Proceedings of the National Academy of Sciences*, 111(25), 9283–
1400 9288. doi:10.1073/pnas.1408296111.

1401 Proekt, A., & Hudson, A. E. (2018). A stochastic basis for neural inertia
1402 in emergence from general anaesthesia. *British Journal of Anaesthesia*,
1403 121(1), 86–94. doi:10.1016/j.bja.2018.02.036.